

# Puppala Madhuvenu

Java Full Stack Developer — Java — Spring Boot

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## Professional Summary

B.Tech Computer Science Graduate (2026) with **CGPA 8.99/10** and hands-on software development experience using **Java, Spring Boot, Hibernate/JPA, REST APIs, MySQL, JavaScript, and React.js**. Experienced in **object-oriented design, data structures, API development, debugging, database optimization, and performance analysis**. Built full-stack and backend applications involving **Spring Security, WebSocket, and concurrent load testing**, including measurable improvements in API latency and throughput. Strong foundation in software engineering, problem solving, and maintainable application development.

## Education

**PBR Visvodaya Institute of Technology and Science**, Kavali

2022 – 2026

B.Tech in Computer Science and Engineering

**CGPA: 8.99/10**

**Sai Cooperative Junior College**, Kavali

2020 – 2022

Intermediate (MPC)

**70.60%**

**AP Model Government School**, Kondapuram

2019 – 2020

SSC

**89.33%**

## Skills

**Programming Languages:** Java, Python, JavaScript, SQL

**Backend:** Spring Boot, Spring Security, JPA/Hibernate, RESTful APIs

**Web Technologies:** HTML5, CSS3, React.js, WebSocket

**Software Engineering:** OOP, Data Structures & Algorithms, SDLC, Debugging, Modular Architecture

**Databases:** MySQL, Relational Database Design, SQL Query Optimization

**Tools:** Git, GitHub, Maven, Postman, VS Code

## Projects

**AI-Powered Intelligent Smart Queue Management System**   **Live Demo**   **GitHub**

May 2026 – Jul 2026

- Architected a **real-time** queue management platform using Java, Spring Boot, MySQL, REST APIs, WebSocket, and **modular architecture**, separating appointment scheduling, token generation, real-time queue management, ETA recalculation, and notification domains.
- Optimized Spring Security authentication by eliminating redundant database lookups and reusing authenticated user context, reducing login latency by **44% (498 ms → 280 ms)**, p95 latency by **48% (1.52 s → 793 ms)**, and increasing throughput by **17% (30 → 35 req/s)** under **100 concurrent users** using k6.
- Validated backend across **authentication, appointment booking, queue management, analytics, and doctor workflows**, sustaining **43.7 req/s**, **26 ms average latency**, **98.6 ms p95 latency**, and **0% request failures** under **100 concurrent users** using k6.
- Reduced analytics API p95 latency by **45% (1.30 s → 711 ms)** and average latency by **58% (419 ms → 177 ms)** at **1,000 concurrent virtual users** by eliminating N+1 database access and unnecessary entity loading while maintaining **0% request failures**.

**Enterprise Employee Management Platform – Java Spring Boot**   **Live Demo**   **GitHub**

Jan 2025 – Mar 2025

- Developed a full-stack employee administration platform using **Java, Spring Boot, Spring Security, Hibernate/JPA, MySQL, REST APIs, and React.js**, implementing employee, department, document, and dashboard workflows.
- Implemented **JWT-based stateless authentication and role-based authorization** for ADMIN, HR, and MANAGER users using Spring Security, including access and refresh token handling and protected REST endpoints.
- Built scalable employee APIs with **server-side pagination, filtering, sorting, JPA Specifications, validation, global exception handling, and audit logging**, improving maintainability and preventing inefficient retrieval of large employee datasets.
- Applied **JPA entity relationships, transactional service-layer operations, document/file management, and Docker-based application packaging** for consistent application deployment and data management.

## Professional Experience

**CodeTantra**

May 2025 – Jun 2025

*Java Full Stack Intern*   **Certificate**

- Implemented hotel booking workflows using **Java, Spring Boot, MySQL, REST APIs, and persistence logic**, translating business requirements into maintainable application components.
- Reduced API latency by **25% (400 ms → 300 ms)** by tracing slow hotel-booking requests to inefficient database operations, optimizing the underlying SQL queries, and refining Spring Boot service-layer processing to eliminate unnecessary work.